

Lesson 12: Power Series Solutions - Frobenius Method

Professor Dave Explains Differential Equations Master Series · Lesson 12

Video: Power Series Solutions Part 2: Frobenius Method **Course:** Applied Ordinary Differential Equations (ENGR 213)

1. Video Intuition & Pedagogical Framing

Professor Dave explains that when an ODE has a singular point (where leading coefficients vanish, like $x^2y'' + xy' + (x^2 - u^2)y = 0$), regular Taylor series fail. If the singularity is 'regular' (tame enough that $xP(x)$ and $x^2Q(x)$ remain analytic), Ferdinand Georg Frobenius showed that multiplying a power series by an unknown fractional or negative power x^r rescues the solution: $y = x^r \sum c_n x^n$. The lowest-order balance yields the Indicial Equation, which dictates the fundamental physics of the solution.

2. Core Theoretical Framework & Rigorous Formulations

The Frobenius Framework at $x_0 = 0$

Given $x^2y'' + x[xP(x)]y' + [x^2Q(x)]y = 0$ with $p_0 = \lim_{x \rightarrow 0} xP(x)$ and $q_0 = \lim_{x \rightarrow 0} x^2Q(x)$:

$$y(x) = \sum_{n=0}^{\infty} c_n x^{n+r}, \quad (c_0 \neq 0)$$

The Indicial Equation (from the lowest power x^r):

$$I(r) = r(r-1) + p_0r + q_0 = 0$$

- **Case 1** ($r_1 - r_2 \notin \mathbb{Z}$): Two clean Frobenius series $y_1 = x^{r_1} \sum a_n x^n$ and $y_2 = x^{r_2} \sum b_n x^n$.
- **Case 2** ($r_1 = r_2 = r$): Second solution requires a logarithm: $y_2 = y_1(x) \ln x + x^r \sum b_n x^n$.
- **Case 3** ($r_1 - r_2 = N \in \mathbb{Z}^+$): $y_2 = Cy_1(x) \ln x + x^{r_2} \sum b_n x^n$ (where C may be zero).

3. Fully Solved Whiteboard Problems

Frobenius Problem: Finding the Indicial Equation and Primary Series

Problem Statement: For $2xy'' + y' + xy = 0$, verify that $x = 0$ is a regular singular point, determine the indicial roots, and formulate the recurrence relation.

Step 1: Verify Regular Singularity

Normalize: $y'' + \frac{1}{2x}y' + \frac{1}{2}y = 0$. $p(x) = xP(x) = \frac{1}{2}$ (analytic). $q(x) = x^2Q(x) = \frac{1}{2}x^2 \rightarrow 0$ as $x \rightarrow 0$ (analytic). Regular singular point confirmed!

Step 2: Derive Indicial Equation

$$r(r-1) + p_0r + q_0 = r(r-1) + \frac{1}{2}r + 0 = r^2 - \frac{1}{2}r = r(r - \frac{1}{2}) = 0.$$

Step 3: Evaluate Indicial Roots

$r_1 = \frac{1}{2}$, $r_2 = 0$. Since $r_1 - r_2 = \frac{1}{2} \notin \mathbb{Z}$, this falls under Frobenius Case 1 (two linearly independent non-logarithmic series!).

Step 4: Recurrence for $r_1 = 1/2$

Substituting $y = \sum c_n x^{n+1/2}$ yields the recurrence $c_n = -\frac{c_{n-2}}{2n(2n+1)}$ for $n \geq 2$, generating alternating even-step coefficients.

⚠ Common Exam Pitfall & Trapdoor Avoidance

Never assume $c_0 = 0$ in a Frobenius problem. The entire method is premised on $c_0 \neq 0$ so that the indicial coefficient accurately isolates the roots r_1, r_2 .

4. Key Takeaways & Exam Strategy Formula Card

🌀 Core Takeaway & Exam Rule

Regular singular points require $y = x^r \sum c_n x^n$. The indicial equation $r(r-1) + p_0r + q_0 = 0$ dictates whether logarithms appear.

- Video reference: [Power Series Solutions Part 2: Frobenius Method](#)
- Always verify boundary conditions and check that candidate solutions satisfy the governing differential equation identically.